



**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

# LESSON PLAN

For

## **Build a Human-AI Workforce with Autonomous AI Agents**

TGS Ref No: TGS-2024043854

Conducted by

**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

**Version 1.3**

## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes                                                                                                                                                                                                                                | Author         |
|----------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1.0            | 12 July 2026              | Initial release — 3-day lesson plan for the autonomous-AI-agent labs (Hermes Agent, OpenClaw, Paperclip).                                                                                                                                                  | Dr. Alfred Ang |
| 1.1            | 14 July 2026              | Revised to a 2-day plan (Day 1: Topics 1 & 2; Day 2: Topic 3 & final assessment). Hermes expanded to 14 labs, OpenClaw to 12, Paperclip to 8 (34 labs total). Learning outcomes aligned to the Course Proposal / TSC; per-lab reference videos added.      | Dr. Alfred Ang |
| 1.2            | 14 July 2026              | Visual redesign: all content slides card-format; YouTube links removed from the deck and Learner Guide (videos remain in the labs); live Hermes screenshots and skill-category grids added; Hermes install lab expanded with the official install methods. | Dr. Alfred Ang |
| 1.3            | 14 July 2026              | Added the masterclass-style Hermes slides (What is Hermes; The Bet: Compounding Value; platform matrix; Top-10 slash commands) and the live Achievements dashboard screenshot to the deck.                                                                 | Dr. Alfred Ang |

## TABLE OF CONTENTS

|                                                          |          |
|----------------------------------------------------------|----------|
| <b>Course Information</b> .....                          | <b>4</b> |
| <b>Learning Outcomes</b> .....                           | <b>4</b> |
| <b>Assessment</b> .....                                  | <b>4</b> |
| <b>Course Schedule</b> .....                             | <b>4</b> |
| Day 1 — Hermes Agent (Topic 1) & OpenClaw (Topic 2)..... | 4        |
| Day 2 — Paperclip (Topic 3) & Final Assessment.....      | 5        |
| <b>Lab Reference (aligned to course topics)</b> .....    | <b>6</b> |

## Course Information

|                             |                                                                            |
|-----------------------------|----------------------------------------------------------------------------|
| <b>Course Title</b>         | Build a Human-AI Workforce with Autonomous AI Agents                       |
| <b>WSQ Course Reference</b> | TGS-2024043854                                                             |
| <b>Training Provider</b>    | Tertiary Infotech Academy Pte Ltd (UEN 201200696W)                         |
| <b>Duration</b>             | 2 days · 8 training hours per day (16 hours)                               |
| <b>Daily Timing</b>         | 9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)          |
| <b>Mode</b>                 | Instructor-led, hands-on autonomous-AI-agent labs (local machine + Docker) |
| <b>Trainer</b>              | Dr. Alfred Ang                                                             |

## Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyse AI (LLM-based) applications across a range of industries to identify their capabilities and limitations.
- LO2: Establish the relationship between AI algorithm/agent design and chatbot/agent efficiency.
- LO3: Evaluate and improve the effectiveness of AI (RAG and agent) applications in product development.

## Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on agent-building tasks that follow the course labs, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

## Course Schedule

### Day 1 — Hermes Agent (Topic 1) & OpenClaw (Topic 2)

| Time       | Duration | Topic / Activity                                                                                                                                                                         |
|------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9:30–9:50  | 20 min   | Welcome, course introduction, ground rules and mandatory digital attendance (AM)                                                                                                         |
| 9:50–10:30 | 40 min   | <b>Topic 1 — Hermes Agent — Your Autonomous Personal Chief-of-Staff: what an autonomous AI agent is, the agent stack (model, memory, skills, tools) and human-in-the-loop governance</b> |

|                    |         |                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |         | <b>(concepts + demo)</b>                                                                                                                                                                                                                                                                                                                             |
| 10:30–11:00        | 30 min  | Hands-on: Lab 1: Install and Setup Hermes; Lab 2: Deployment — Local Install & Hermes Desktop App; Lab 3: Memory and Plugins                                                                                                                                                                                                                         |
| 11:00–11:15        | 15 min  | Tea break                                                                                                                                                                                                                                                                                                                                            |
| 11:15–13:00        | 105 min | Hands-on: Lab 4: Skills; Lab 5: Providers & Model; Lab 6: MCP and Tools; Lab 7: Cron Jobs & Automation; Lab 8: Subagents & Delegation; Lab 9: Profile and Kanban                                                                                                                                                                                     |
| 13:00–14:00        | 60 min  | Lunch break                                                                                                                                                                                                                                                                                                                                          |
| 14:00–14:40        | 40 min  | Hands-on: Lab 10: Security; Lab 11: Use Case — Make a Video with Hyperframe; Lab 12: Visualization; Lab 13: Build a Multi-Agent Workflow; Lab 14: Webhook                                                                                                                                                                                            |
| <b>14:40–15:20</b> | 40 min  | <b>Topic 2 — OpenClaw — Automating a Business Back-Office: the OpenClaw gateway daemon, providers, channels, skills, tools, dreaming and crons for a business back-office (concepts + demo)</b>                                                                                                                                                      |
| 15:20–15:35        | 15 min  | Tea break                                                                                                                                                                                                                                                                                                                                            |
| 15:35–18:15        | 160 min | Hands-on: Lab 15: Install OpenClaw; Lab 16: Models & Providers; Lab 17: Channels; Lab 18: Skills; Lab 19: Tools & Integrations; Lab 20: Commands; Lab 21: Cron Jobs & Heartbeat; Lab 22: Memory & Context; Lab 23: Security; Lab 24: Multi-Agent Workforce; Lab 25: Dreaming — Idle Reflection & Self-Improvement; Lab 26: Use Cases & Key Functions |
| 18:15–18:30        | 15 min  | Day 1 recap, Q&A and PM digital attendance                                                                                                                                                                                                                                                                                                           |

*Total training time: 480 minutes (8 hours).*

## Day 2 — Paperclip (Topic 3) & Final Assessment

| Time              | Duration | Topic / Activity                                                                                                                                                                    |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9:30–9:45         | 15 min   | Day 1 recap and mandatory digital attendance (AM)                                                                                                                                   |
| <b>9:45–10:30</b> | 45 min   | <b>Topic 3 — Paperclip — Running &amp; Governing a Company of AI Agents: Docker-hosted company of agents, connecting OpenAI Codex, human approval gates, task tracking, budgets</b> |

|             |         |                                                                                                                                         |
|-------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------|
|             |         | <b>and audit (concepts + demo)</b>                                                                                                      |
| 10:30–11:00 | 30 min  | Hands-on: Lab 27: Setup Paperclip; Lab 28: Connect OpenAI Codex to Paperclip                                                            |
| 11:00–11:15 | 15 min  | Tea break                                                                                                                               |
| 11:15–13:00 | 105 min | Hands-on: Lab 29: Configure Paperclip; Lab 30: Track AI Tasks; Lab 31: Automate AI Hiring; Lab 32: Setup AI Agents                      |
| 13:00–14:00 | 60 min  | Lunch break                                                                                                                             |
| 14:00–15:30 | 90 min  | Hands-on: Lab 33: Security; Lab 34: Assign Task. Course recap and next steps                                                            |
| 15:30–15:45 | 15 min  | Tea break                                                                                                                               |
| 15:45–16:15 | 30 min  | Consolidation, Q&A and assessment preparation                                                                                           |
| 16:15–16:30 | 15 min  | <b>Briefing for Assessment</b>                                                                                                          |
| 16:30–17:30 | 60 min  | <b>Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book</b>                                                        |
| 17:30–18:30 | 60 min  | <b>Practical Performance (PP) — hands-on agent-building tasks that follow the course labs, 1 hour, open book. PM digital attendance</b> |

Total training time: 480 minutes (8 hours).

## Lab Reference (aligned to course topics)

| Topic / Platform                                                            | Weighting | Labs                                                                                                  |
|-----------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------|
| <b>Topic 01: Hermes Agent — Your Autonomous Personal Chief-of-Staff</b>     | 33%       | Lab 1, Lab 2, Lab 3, Lab 4, Lab 5, Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13, Lab 14 |
| <b>Topic 02: OpenClaw — Automating a Business Back-Office</b>               | 33%       | Lab 15, Lab 16, Lab 17, Lab 18, Lab 19, Lab 20, Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26        |
| <b>Topic 03: Paperclip — Running &amp; Governing a Company of AI Agents</b> | 34%       | Lab 27, Lab 28, Lab 29, Lab 30, Lab 31, Lab 32, Lab 33, Lab 34                                        |